

Question	Answer	Marks
7(a)(i)	$P = I_0^2 R$	A1
7(a)(ii)	$P = 4I_0^2 R$	A1
7(b)	sketch: square wave of period T , with P always non-zero	B1
	horizontal lines, from 0 to $0.5T$ and from $1.0T$ to $1.5T$, all at the same level that the scale indicates to be $I_0^2 R$	B1
	horizontal lines, from $0.5T$ to $1.0T$ and from $1.5T$ to $2.0T$, at a level that is four times higher than the lower lines	B1
7(c)(i)	$\langle P \rangle = (5/2)I_0^2 R$	A1
7(c)(ii)	$\langle P \rangle = I_{\text{r.m.s.}}^2 R$	C1
	$I_{\text{r.m.s.}}^2 R = (5/2)I_0^2 R$	A1
	$I_{\text{r.m.s.}} = \sqrt{(5/2)} I_0$	

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2(c)(i)	$5/2$	M1